SPECIFIC CONDITIONS

A. General ***Conditions***

1 Wiring

1 Wiring shall conform to IS 732 wiring shall not cross room through roof slab.
2 Light wiring and power wiring shall be segregated.
3 The points of connection in wires shall be properly soldered.
4 If different materials happen to come at a connection point, corrosion inhabiting compound shall be applied.
5 Bare copper conductors which came close to live exposed portions inside switch boards/DB etc. shall be sleeved and connected using suitable sockets.
6 Wires and copper conductors which come inside switch boards/DB etc. shall be properly clipped with suitable ties, neatly arranged and dressed before giving connections.

2 Fixing Conduits.

a) PVC pipes shall be clamped with GI saddles/painted MS saddles/PVC spacer saddles, spacing of clamps shall not exceed 60cms. For fixing saddles clips, minimum 30mm MS screws shall be used. Fibre plug/raw plug/Phil plug alone will be permitted to fix screws and holes for providing plugs are to be drilled properly. Under no circumstances wooden pegs will be permitted.
b) For fixing conduits in recess, grooves to the required depth shall be cut and pipes saddled in the grooves as is done for camping on surface. It can be fixed with staples also. In either case the spacing between saddles clips/staples shall not exceed 60cms. After fixing conduits as described the groove is to be filled and plastered with 1:6 Cement Mortar and the surface finely finished.
c) All the junction boxes hall be screwed and fixed to the wall/ceiling.
d) Conduits shall be fixed with saddles at distance not exceeding 30cm from a conduit accessory.
e) The size of conduits shall be so fixed as to hold number of wires as stipulated in IS 732.
f) Layout of the conduits shall be so arranged so as not to disturb the aesthetic view of the building.
g) Male/Female adopters shall be provided for the proper termination of PVC conduits if they are in open position.

3 Fixing Switch Boards.

a) Normally switch boards shall be fixed at a height of 1.35m from ground level. It can be lowered to a height of not less than 30cm from the ground level. The position of the switch board shall be fixed according to the directions of the field officers concerned.
b) M.S. Boxes shall be fabricated from 16 SWG MS sheet duly painted with 2 coats of synthetic enamel paint over a cot of red oxide primer. The front side shall be fold and provided with taped holes for holding screws. Necessary knockouts shall be provided for the entry of conduits. 25x3mm brass bolt with nut shall be provided before they are fixed. Where M.S. Boxes are fixed on surface, they shall be fixed with adequate No. of 30mm M.S. screws.
c) Front covers with 3mm, thick hylam sheet are permitted for switch boards of size only up to 300x200mm for switch boards of size 375x300mm and above 6mm thick hylam sheets shall be used. The hylam sheets shall be screwed to the box using 20mm steel/OD. Plated screws with cup washers.
d) Screws shall be provided in variably at all corners while fixing hylam sheet cover. Under no circumstances, the spacing between adjacent screws shall exceed 150mm.

4 Fixing Accessories.

a) Accessories such as switches, plug sockets, fan regulators etc. are to be arranged on hylam sheet cover in such a manner that there shall be a clear space of 25mm along the periphery on all sides. There shall be a clear space of at least 6mm in between individual switches/sockets etc. as the case may be. The spacing between adjacent fan regulators and between regulators and switches/plug sockets shall be 25mm. Size of the switch box shall so selected so as to have this spacing when the accessories are arranged.
b) Only one row of switch/plug sockets shall be arranged on a switch board.
c) The accessories shall be rigidly fixed to the hylam sheet using suitable size nut and bolt. d) Ceiling rose can be screwed directly to conduit outlets.

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5 Fixing lamps/fittings/call bells/ fans etc.

a) Batten holders can be fixed directly to conduit outlets.

b) Bulk head fittings can be fixed either in recess or surface using 30mm iron screws. c) Mirror lamp can be fixed directly on wall using 30mm screws.

d) Bracket fittings can be fixed on MS boxes with hylam sheet cover. The MS box as well as the fittings shall be earthed. Bracket shall be fitted using bolt and nut.

e) Round block for suspending fluorescent fitting shall be screwed using 2 nos.. 50mm M.S. screws.

f) T.W. reaper for fixing fluorescent fittings shall be fixed using at least 3 Nos. of 50mm M.S. screw and the fittings shall be screwed to the reaper using 20mm MS screws.

g) Fan clamps to be fitted as per ISS.

6 Installation of DBs.

a) DBs shall normally be fixed at a height of 1.5m from ground level. Two DBs shall not be at a distance less than 2m.

7 Cabling.

a) Cables shall conform to IS 1554 and shall have ISI marking.

b) Cables shall be laid conforming to relevant IS specification.

c) At points of road crossing, cables are to be drawn through GI pipes laid at a depth of 1m.

8 Panel boards/meter boards.

a) Smaller size panel boards shall be fabricated using 1.6mm MS sheet and bigger size boards using 2.0mm MS Sheet.

b) Separate space for mounting metering equipment to and indicators are to be provided.

9 Earthing.

a) Earthing shall conform to IS 3043/1987.

10 Identification.

a) Identification letters shall be written legibly in English language with white enamel paint on switch boards up to DBs cable.

11 **General Requirements.**

a) The contractor shall conduct at site tests and furnish test results as specified in IS 732. b) As fitted drawing shall be submitted (3 copies) to the department.

c) He shall prepare schematic diagrams, plans etc. required for submission to electrical inspectorate. He shall also follow up with electrical inspectorate and KSEB for getting the installations energized immediately on completion of work.

d) The contract's supervisor shall be at site.

B) Special Conditions.

1 The project is a time bound one. The selected contractor shall complete the work in all respects according to the time schedule appended herewith maintaining top quality of work.

2 The contractor shall produce samples of all materials and get the confirmation from the departmental officers concerned before starting the work.

3 As far as possible, same brand of products shall be used for the entire works (which have been approved. If, under any circumstances, the contractor intends to change the brand prior approval in writing shall be obtained from the departmental officers)

4. As per GO (MS) No. 73/2013/PWD dated 31.08.2013 the defect liability period of the electrical works will be as follows:

a. Original works - 36 Months
b. Special repairs - 18 Months
c. Ordinary repairs and maintenance - 6 Months

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GOVERNMENT OF KERALA

Abstract

PWD - Time of Completion of works and Penalties/Liquidate Damages - Provisions in the Kerala Public Works Department Manual & Standard Bid Document - Modified - Orders issued.

PUBLIC WORKS(H)DEPARTMENT

G.O.(Ms)No.21/2024/PWD Dated,Thiruvananthapuram, 02-04-2024

Read: 1. G.O(P)No.13/2012/PWD dated 01.02.2012

2. G.O(P)No.3/2017/PWD dated 09.08.2017

3. Letter No.CEPWD/9127/2021/Admn/PLA dated 24.09.2022 from the Chief Engineer(Administration), PWD.

ORDER

In clause 2112.1 of the Govt. Order read 1st paper above(Kerala PWD Manual), the fines for extension of time of completion of work is stipulated as follows:

Period	***Rate of Fine***
First Extension	*1% of the PAC subject to a minimum of Rs.1000/- and maximum of Rs.50,000/-*
Beyond First Extension	*2% of the PAC subject to the a minimum of Rs.2000/- and maximum of Rs.1,00,000/-*

Clause 2116.1 of the KPWD Manual is as follows:

"If the contractor fails to maintain the required progress as per conditions of contractor to complete the work and clear the site on or before the contract or extended date of completion, he shall, without prejudice to any other right or remedy available under the law to the Employer on account of such breach, pay to the employer as liquidated damages an amount calculated 0.1% of the

contract price of the work for every work of delay subject to a maximum of 10% of the contract price."

2) As per clause 15.2 of the Govt.Order read 2 nd paper above (Standard Bid Document),

The Agreement Authority if satisfied , that the work can be completed by the Contractor within a reasonable time after the specified time of completion, may allow further extension of time at its discretion as per clause 19. In the event of extension granted being with Liquidated Damages, the Employer will be entitled without prejudice to any other right or remedy available in that behalf, to recover from the Contractor as agreed damages equivalent to 1(one)% of the contract price per week of delay or part thereof.

3) As there exis ts a disparity in the above two clauses Govt. have called for proposal from the Chief Engineer(Admn), PWD. The proposal submitted by the Chief Engineer(Admn), as per letter read above has been examined in detail. In order to have more clarity and also to ensure timely completion of the works, Govt. are pleased to modify the provisions *of* KPWD Manual and Standard Bid Document(SBD) as follows:

(i) Penalties @ 0.1% of the agreed Probable Amount of Contract (PAC) shall be imposed as liquidated damages for the delay for each week beyond the scheduled date of completion subject to a maximum of 10% of the agreed PAC for the reasons attributable to the contractor.

(ii) Maximum extension for time of completion that can be granted for a work by the Agreement Authority without any penalty shall be limited to the half of the original ToC.

Prior approval of next higher authority is required for granting time extension

without penalty beyond half of the original ToC. ToC shall be calculated using technically available tools like Bar chart, Gant chart, Critical Path Method, etc. The mile stones and liquidated damages shall be fixed based on this.

4) The Chief Engineer(Administration), PWD shall make necessary modification in KPWD Manual and SBD accordingly.

(By order of the Governor)
K BIJU I A S
SECRETARY

To

All the Chief Engineers, PWD,Thiruvananthapuram

The CEO, KRFB, Thiruvananthapuram

The PD, PMU-KRFB, Thiruvananthapuram

The Managing Director, RBDCK/RICK Ltd./KSCC

The Principal Accountant General (Audit), Kerala,Thiruvananthapuram.

The Finance Department (I&PW-B2/47/2022-FIN dated 22.03.2023)

The Local Self Govt Department (U.O File No.LSGD-AC2/223/2015)

The Water Resources Department

The I&PR(Web&New Media)Department

Stock File/Office Copy(H3/67/2021/PWD)

Forwarded /By order,

Section Officer.